

Beaconhouse National University



Assignment: Reading 2.2

Subject: Functional English

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How do astronomers know the age of the planets and stars?

Determining the age is crucial in understanding a planet's or star's formation and evolution especially when searching for signs of life on other planets. It's a difficult and complex task because Stars, like our sun, maintain consistent characteristics like brightness, temperature and size for over billions of years and planets properties such as temperature are often influenced by the stars as they orbit.

To suss out a star's age, astronomers use various methods like observing the brightness and color changes over time then comparing spin speeds to mathematical models which is called gyrochronology after which they track declining magnetic activity and using asteroseismology to analyze surface vibrations. These techniques collectively estimate stars' ages, with our sun estimated at 4.58 billion years.

My Views

I never imagined that there would be something like calculating a star's age. This was a very informative piece and opened my mind to so many other possibilities of space.

My Questions

Why is it essential to determine the age of a planet or a star?

The Earth needs the moon for various reasons like ocean tides, calendars and day and night etc but why do planets like Jupiter require moons, and why do they have multiple moons instead of just one?

Sources

<https://www.space.com/how-do-astronomers-know-age-of-planets-stars>